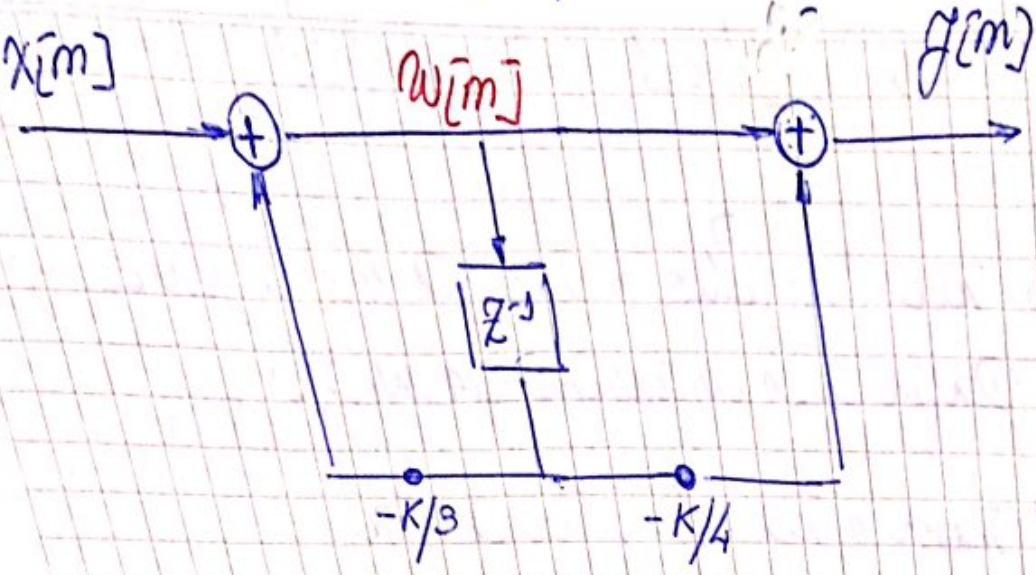
9 DE FEBRERO DE 2015

1)



a) TENER $\longrightarrow \boxed{z^{-1}} \longrightarrow \equiv \longrightarrow \boxed{\delta[m-1]} \longrightarrow$

• $w[m] = x[m] + \left\{-\frac{K}{3}\right\} w[m] * \delta[m-1]$

$w[m] = x[m] - \frac{K}{3} w[m-1]$ (1)

• $y[m] = w[m] - \frac{K}{4} w[m-1]$ (2)

DE (1) $W(z) = X(z) - \frac{K}{3} z^{-1} W(z)$

$$W(z) = \frac{X(z)}{1 + \frac{K}{3} z^{-1}} \quad (3)$$

DE (2) $Y(z) = W(z)\left[1 - \frac{K}{4} z^{-1}\right]$ (4)

(3) EN (4) $Y(z) = X(z) \frac{1 - \frac{K}{4} z^{-1}}{1 + \frac{K}{3} z^{-1}}$

$\Rightarrow H(z) = \frac{1 - K/4\, z^{-1}}{1 + K/3\, z^{-1}}$; ROC: $z > -K/3$ / $|z| > |K|/3$

POR SER CAUSAL!

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b) PARA SER ESTABLE: $\max |p_k| < 1$.

$$\Rightarrow \left|\frac{K}{3}\right| < 1 \Rightarrow |K| < 3.$$

FASE MÍNIMA $\Rightarrow$ TODOS LOS POLOS Y CEROS DE $H(z)$ DEBEN ESTAR DENTRO DE LA CIRCUNFERENCIA UNIDAD.

- CON $|K| < 3$ TENGO A TODOS LOS POLOS DENTRO
- CERO: EN $z = K/4$ Y SABIENDO QUE $|K| < 3$.

$\Rightarrow$ EL CERO SIEMPRE ESTARÁ DENTRO.

$\Rightarrow |K| < 3 \rightarrow H(z)$ ESTABLE

$\rightarrow$ FASE MÍNIMA.

c) PARA $K = 4$

$$H(z) = \frac{1 - z^{-1}}{1 + \frac{4}{3} z^{-1}} = \frac{z - 1}{z + 4/3} \quad ; \quad |z| > 4/3$$

$$H(z) = \frac{1}{1 + 4/3\, z^{-1}} - \frac{z^{-1}}{1 + 4/3\, z^{-1}}$$

$$h[m] = \left(-\frac{4}{3}\right)^m u[m] - \left(-\frac{4}{3}\right)^{m-1} u[m-1]$$

$\rightarrow$ NO ES ESTABLE

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$g[m] = x[m] * h[m]$

RESPUESTA AL ESCALON: $x[m] = u[m]$

$$Y(z) = X(z)\,H(z) \quad ; \quad u[m] \longrightarrow \frac{1}{1-z^{-1}} \; ; \; |z|>1$$

$$Y(z) = \frac{1}{1-z^{-1}} \; \frac{1-z^{-1}}{1+\frac{4}{3}z^{-1}}$$

$$Y(z) = \frac{1}{1+\frac{4}{3}z^{-1}} \quad ; \quad |z| > 4/3$$

$$g[m] = \left(-\frac{4}{3}\right)^m u[m]$$

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$$g[m] = \underbrace{x[2m]}_{g_1[m]} + j\,\underbrace{x[2m+1]}_{g_2[m]}$$

3)

$$X[K] = \sum_{m=0}^{N-1} x[m]\, W_N^{Km}$$

$$X[K] = \sum_{m=0}^{N/2-1} x[2m]\, W_N^{2mK} + \sum_{m=0}^{N/2-1} x[2m+1]\, W_N^{(2m+1)K}$$

$$X[K] = \sum_{m=0}^{N/2-1} x[m]\, W_{N/2}^{mK} + \sum_{m=0}^{N/2-1} x[2m+1]\, W_{N/2}^{mK}\, W_N^{K}$$

$$X[K] = \sum_{m=0}^{N/2-1} x[2m]\, W_{N/2}^{mK} + W_N^{K} \sum_{m=0}^{N/2-1} x[2m+1]\, W_{N/2}^{mK}$$

$$X[K] = Y_1[K] + W_N^{K}\, Y_2[K], \quad K = 0, \dots, N-1$$

PERO: $W_N^{K+N/2} = -W_N^{K}$

LUEGO:

$$\begin{cases} X[K] = Y_1[K] + W_N^{K}\, Y_2[K] & (1) \\ X[K+\frac{N}{2}] = Y_1[K] - W_N^{K}\, Y_2[K] & (2) \\ K = 0, \dots, N/2 - 1 \end{cases}$$

(1)+(2) $\quad Y_1[K] = \dfrac{X[K] + X[K+N/2]}{2}$

(1)-(2) $\quad Y_2[K] = \dfrac{X[K] - X[K+N/2]}{2W_N^{K}}$

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$$\Rightarrow Y[K] = Y_1[K] + j\,Y_2[K]$$

$$Y[K] = \frac{X[K] + X[K+N/2]}{2} + j\,\frac{X[K] - X[K+N/2]}{2W_N^K}$$

b) PARA $K=0$.

$$Y[K] = \frac{X[0] + X[N/2]}{2} + j\,\frac{X[0] - X[N/2]}{2}$$

$$Y[K] = \frac{X[0] + X[7]}{2} + j\,\frac{X[0] - X[7]}{2}$$

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